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Proof of Stake

The Tech
Behind Crypto

Definition & Core Idea

Proof of Stake (PoS) is a consensus mechanism where the right to propose and validate the next block is given probabilistically in proportion to the amount of cryptocurrency a participant locks ("stakes") in the protocol.

Definition & Core Idea

- In contrast to PoW's race of computation, PoS runs a lottery of economic commitment: more stake = higher odds, not a guarantee.
- Security is economic: if a validator cheats, part/all of their stake can be slashed (destroyed).

Mini Example

If the total stake in a network is 10,000 coins and you stake 500 coins (5% of total), your baseline chance of being selected to propose a block in a given round is ~5% (actual probability depends on the specific PoS design).

End-to End Workflow

- Stake/Lock – A user locks some coins in a validator (own node) or delegates to a stake pool.
- Random Selection – The protocol runs a pseudo-random selection (often via a Verifiable Random Function, committee sampling, or epoch/slot scheduling) to choose a block proposer and a set of validators who will attest/vote.

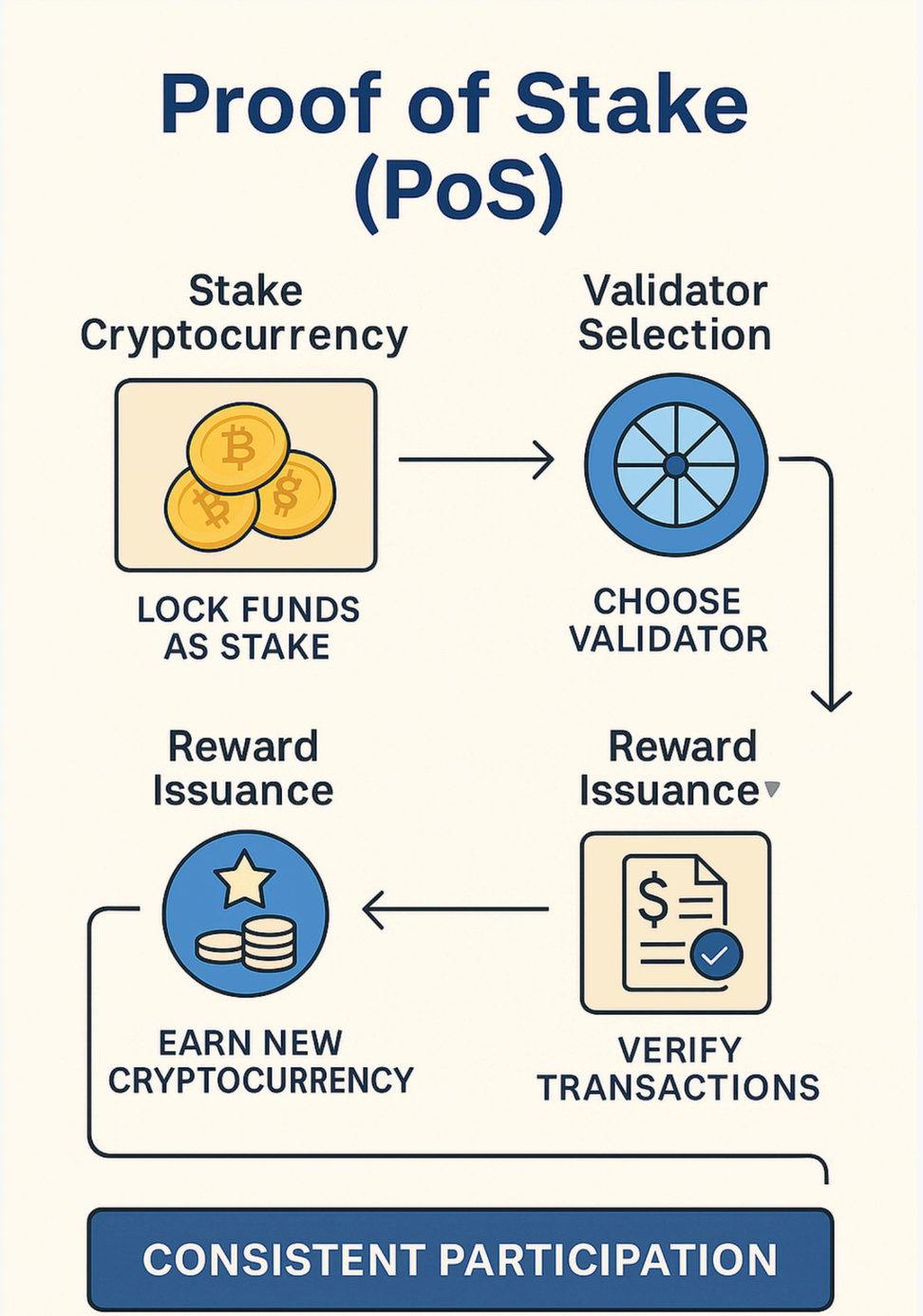
End-to End Workflow

- **Block Proposal – The chosen validator builds a block from pending transactions and broadcasts it.**
- **Attestation/Voting – A committee of validators attests that the block is valid. Different PoS families do this differently (BFT voting, attestations over epochs, etc.).**

End-to End Workflow

- **Finality – After enough attestations (e.g., ≥2/3 of stake in BFT-style PoS), the block becomes final (hard to revert).**
- **Rewards & Penalties – Honest validators receive rewards/fees; misbehavior (double-signing, censorship, being offline during finality failure) triggers slashing or inactivity leak.**

End-to End Workflow



Worked Miniature Schedule

- Epoch = 32 slots (e.g., ~6–12 minutes depending on chain).
- In each slot, one proposer is selected; a committee of attesters votes.
- If ≥2/3 of stake attests across consecutive checkpoints, the epoch is finalized.

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Worked Miniature Schedule

In most PoS systems (like Ethereum 2.0), time is divided into slots and epochs.

Worked Miniature Schedule

1. Slot = a chance to create a block
   - A slot is a short time window (e.g., 12 seconds in Ethereum).
   - In each slot:
     - One validator is randomly chosen as the proposer (block creator).
     - A group of validators (called a committee) is chosen to attest (vote) on the block.

Worked Miniature Schedule

📌 Think of a slot like one student getting to present in class while a group checks if the answer is correct.

Worked Miniature Schedule

2. Epoch = a batch of slots
   - An epoch is a collection of slots (e.g., 32 slots in Ethereum).
   - At the end of an epoch, if enough validators (≥ 2/3 of the stake) have voted correctly, the blocks in that epoch are considered finalized.
   - Finalized means: You can’t go back and change them (like submitting final exam papers).

Worked Miniature Schedule

📌 Think of an epoch as a full class session with 32 presentations, and at the end the teacher stamps the record book saying “all presentations confirmed”.

Worked Miniature Schedule

Rewards & Penalties

- Validators who do their job (propose/attest correctly) get rewards.
- Validators who are offline or dishonest get penalties/slashing.

Worked Miniature Schedule

📌 Like students: attend + answer correctly = marks.
Absent or cheat = marks cut.

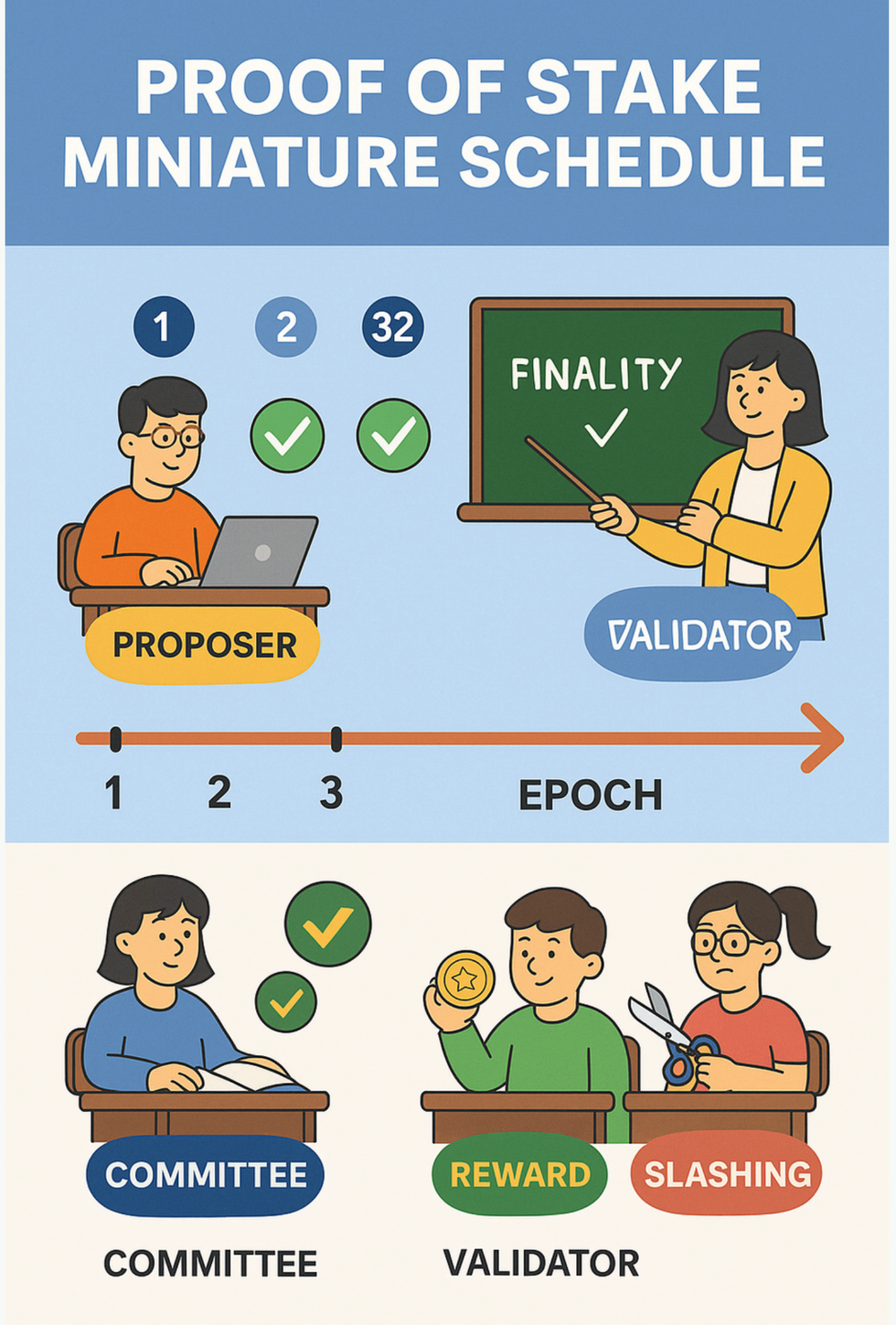
Worked Miniature Schedule

Imagine a class of 32 students (representing 32 slots = 1 epoch).

- Each minute (slot), one student is chosen to present (propose block).
- A group of classmates checks and says ✅ or ❌ (attestations).
- After all 32 are done, the teacher signs the record (finality).
- Honest students get chocolates (rewards 🍫). Cheaters lose stars (slashing ✂️).

This cycle repeats every epoch.

Worked Miniature Schedule



Major Variants

- Chain-based PoS (probabilistic lottery): The protocol picks a single proposer per slot with probability ∝ stake; others verify.
- Intuition: closest to the “lottery tickets” analogy.
- BFT-style PoS (e.g., Tendermint/HotStuff family): A rotating proposer suggests a block and a committee votes; block is final when ≥2/3 of stake signs.
- Benefit: near-instant finality once a quorum is reached.

Major Variants

- Delegated PoS (DPoS): Token holders elect a small set of delegates/validators (e.g., 21–100) who take turns producing blocks.
- Benefit: high throughput; trade-off: higher governance centralization.
- Nominated PoS / Stake Pools: Holders nominate trusted validators (Polkadot), or join stake pools (Cardano).
- Benefit: lets small holders participate and earn.

Security & Economics

- 51% stake attack (analysis): An attacker must purchase/control >50% of total stake, which is economically massive and pushes the asset price up while exposing the attacker to slashing and market loss if detected.

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Numerical Intitution

If total stake is ₹1000 crore worth of coins, a 51% attacker needs ₹510+ crore at rising prices and risks losing a large fraction to slashing. That’s a deterrent built into protocol economics.

Known Challenges

“Nothing-at-Stake” problem

- Issue: In PoS it’s cheap to vote on multiple forks.
- Mitigation: Slashing for double-signing + finality gadgets (e.g., checkpoints, epochs).

Known Challenges

Stake centralization / rich-get-richer

- **Issue: Large stakers may win more often.**
- **Mitigation: Stake pools, delegation, reward curves/caps (Cardano's k-parameter encourages many pools), randomness beacons and rotation.**

Known Challenges

Long-range attacks

- Issue: Very old keys try to rewrite ancient history.
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- Mitigation: Weak subjectivity checkpoints—clients must occasionally trust a recent finalized checkpoint; finality makes deep rewrites infeasible.

Known Challenges

Liveness under partitions

- **Issue: Network splits can stall finality.**
- **Mitigation: BFT quorum rules; inactivity leaks; fork-choice rules (e.g., LMD-GHOST in Ethereum) to converge after partition heals.**

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Thank You For Your Attention